STRATEGIC ELECTRONICS DIVISION

BIST Embedded Telemetry Board

Reliability & Fault-Tolerance in Defense Systems

Presentation Agenda

System Architecture

Core design and hardware layout.

BIST Implementation

Self-test logic & fault detection.

Telemetry Usage

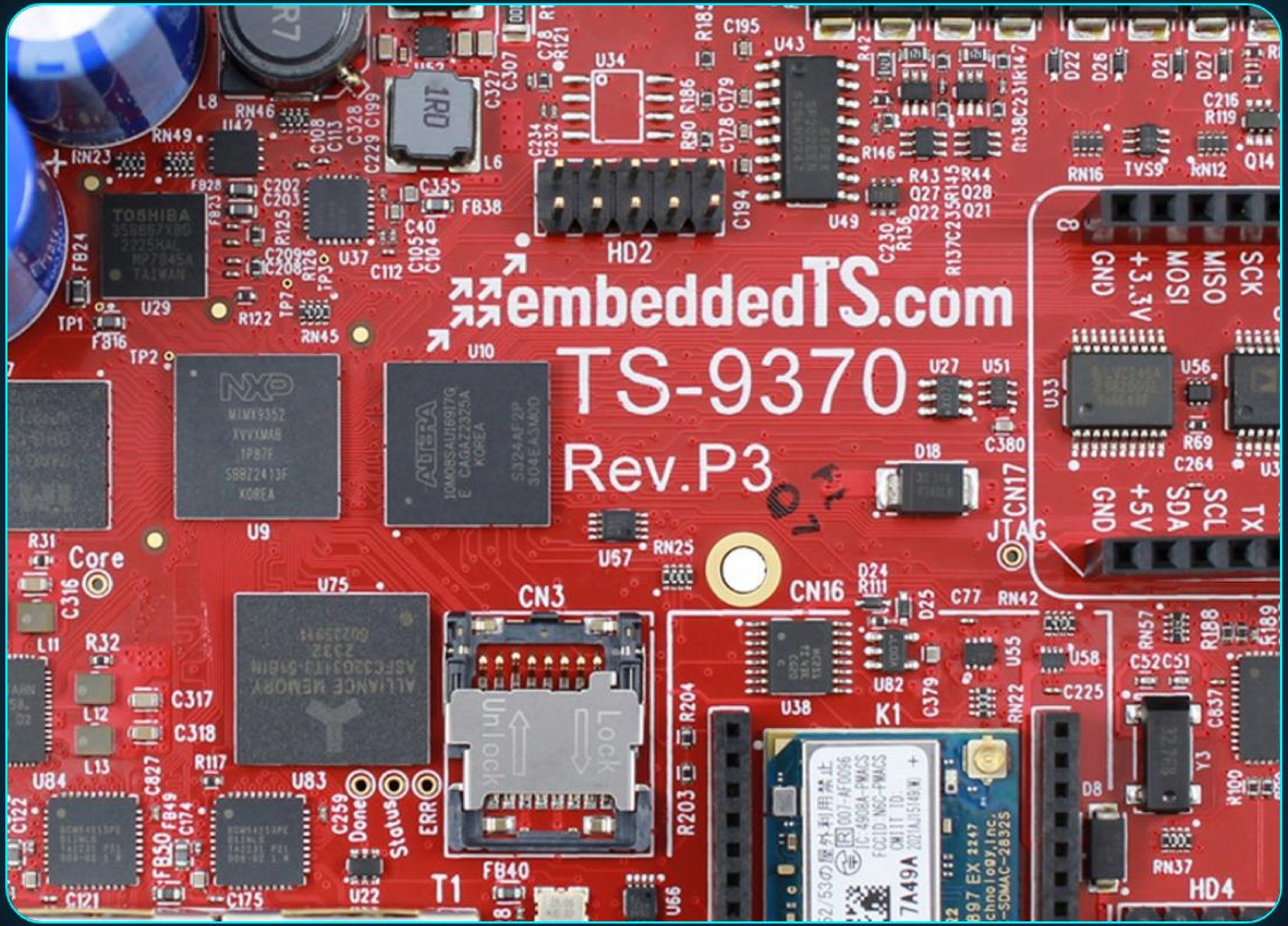
Real-time data acquisition.

Telemetry Systems in Defense

Mission Critical Monitoring

Telemetry involves the automatic measurement and transmission of data from remote sources to IT systems for analysis. In defense, this ensures:

- Real-time health monitoring of flight vehicles.
- Performance validation under extreme G-loads.
- Synchronized sensor fusion for flight control.



Why Built-In Self-Test?

99.9%

System Availability

Core Objectives

BIST allows the hardware to test itself without requiring expensive external Test Equipment (ATE).

- **Autonomous Diagnostics:** Reduces maintenance time.
- **On-line Monitoring:** Detects faults during mission.
- **Reliability:** Guaranteed performance in unreachable environments.

High-Level System Architecture

The Telemetry Board integrates acquisition, processing, and self-test logic in a single compact module.

Internal BIST Architecture

TPG (Test Pattern Gen) → CUT (Circuit Under Test) → ORA (Response Analyzer)

TPG: Generates pseudo-random test vectors to exercise the internal logic paths.

ORA: Compares output signals with expected 'Golden Signatures' to verify integrity.

FPGA: The Brain of the System

Xilinx/Altera Core

Usage: Implements the entire BIST controller logic and handles high-speed data stream formatting.

- Parallel processing for real-time telemetry.
- Reconfigurable logic for adaptive self-tests.
- Manages Sensor-to-RF timing synchronization.

Test Pattern Generator (TPG)

Definition

Hardware block responsible for generating stimuli vectors to test the internal circuitry paths.

Project Usage

Generates 2,000+ unique patterns during startup to verify logic integrity before launch.

TPG Core: LFSR Technology

Pseudo-Random Logic

The Linear Feedback Shift Register (LFSR) is used for exhaustive testing with minimal hardware overhead.

$$f(x) = 1 + x^3 + x^7$$

This polynomial defines the feedback taps, ensuring a maximum length sequence of test patterns that cover all possible logic states.

Output Response Analyzer (ORA)

The ORA compresses the results of thousands of tests into a single **Signature**.

Signature Match

A mismatch between the generated signature and the 'Golden Reference' triggers an **Instant Fault Alarm**.

MISR Technique

The project uses Multiple Input Signature Registers (MISR) to analyze parallel data buses simultaneously.

Sensor Interface & Front-End

Sensor Module	Measured Parameter	BIST Verification
MEMS Accel	Flight G-Load	Offset Calibration Check
Thermistor Array	Board Temperature	Continuity & Range Test
Pressure Transducer	Altitude/Velocity	Static Level Comparison

Analog to Digital Converters

ADS1256 High Speed

Usage: Converts analog sensor inputs into 24-bit digital packets for the FPGA processing core.

BIST Role: An internal loopback test verifies the DAC-to-ADC path to ensure no quantization errors.

Sensors

Memory & Data Buffering

Flash Storage

Stores the "Golden Signatures" and logs historical BIST results for post-flight analysis.

SRAM Buffer

High-speed temporary cache for real-time data packets before RF transmission.

Real-time Diagnostics Cycle

Power On &
Initial Self-Test

Sensors Data
Acquisition

Concurrent BIST
Monitoring

RF Telemetry
Uplink Transmission

RF Communication

RF Uplink & Communication

The board utilizes a S-Band RF transmitter for mission data uplink.

- **Data Rate:** Up to 10 Mbps telemetry link.
- **Protocol:** PCM/FM Encoding.
- **Range:** Verified for over 200km flight profiles.

Power Distribution & Monitoring

PMIC Module

Regulates 28V DC into 3.3V, 1.8V, and 1.2V rails for FPGA and Analog chips.

BIST Integration

Real-time current and voltage monitoring to detect shorts or component aging.

Reliability & Fault Coverage

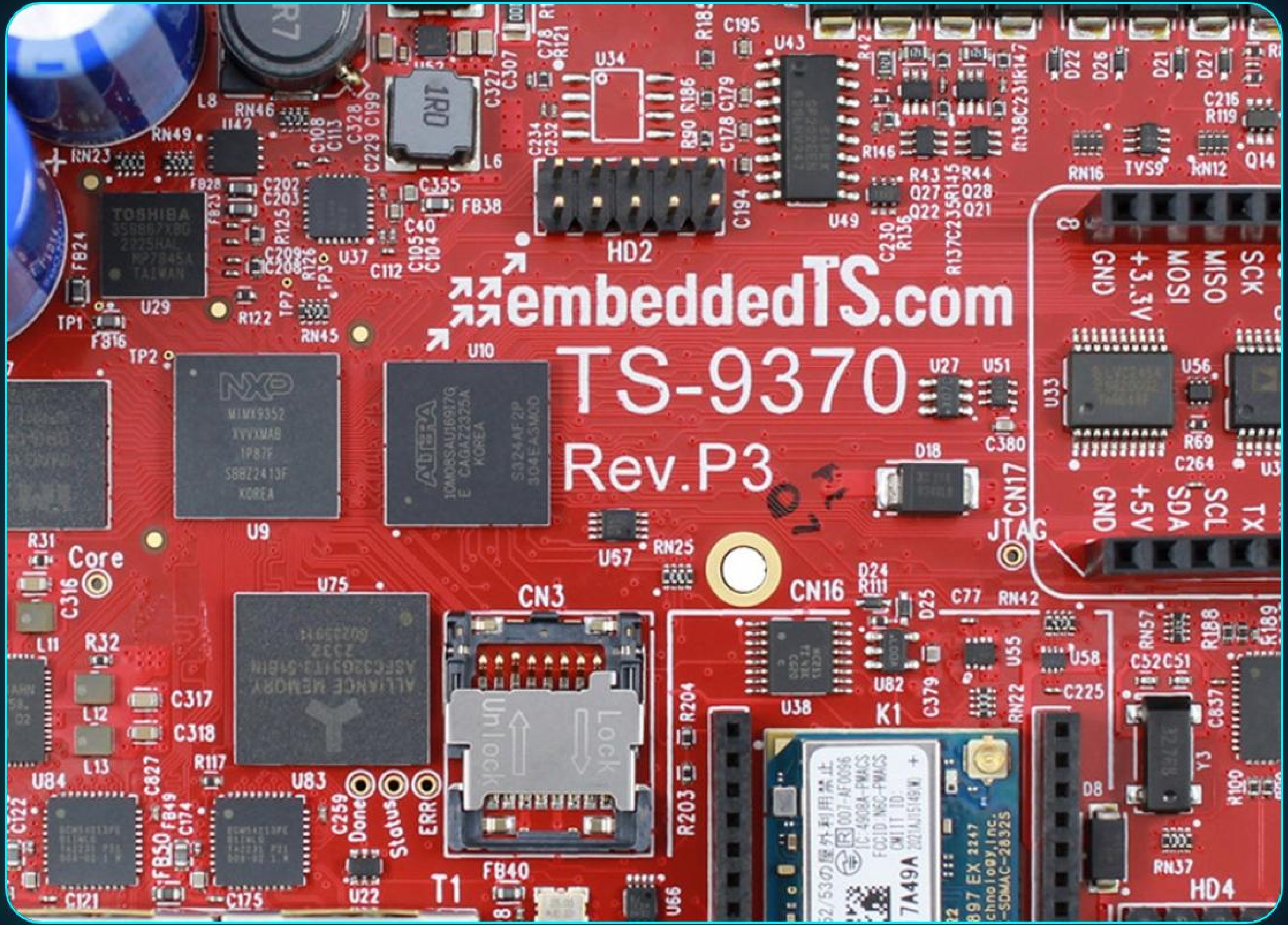
Test Metrics

The project achieves high-level verification through exhaustive BIST patterns.

MIL-STD-883

Designed to comply with high-reliability standards for defense aerospace electronics.

BEL Project Case Study



UAV Telemetry Implementation

Deployed in multi-role drones for real-time mission parameters and structural health monitoring.

- Zero failures reported in flight.
- Automatic pre-flight clearing via BIST in under 500ms.

Key Project Advantages

Reduced Cost

No need for massive external testing infrastructure at site.

Fast Turnaround

Diagnostic results available within seconds for ground teams.

Enhanced Safety

Early warning for internal hardware degradation.

Thank You

Built-In Self-Test Embedded Telemetry Board

Strategic Electronics Division | Bharat Electronics Limited

Contact: technical-support@bel-india.com

Image Sources

https://www.logo.wine/a/logo/Bharat_Electronics_Limited/Bharat_Electronics_Limited-Logo.wine.svg

Source: www.logo.wine

Thumbnail for www.army-technology.com

https://www.army-technology.com/wp-content/uploads/sites/3/2025/08/embeddedTS-Lead-Image.jpg

Source: www.army-technology.com

https://wevolver-project-images.s3.us-west-1.amazonaws.com/0.q33m6fx0fodxilinx-virtex-5-fpga-chip-circuit-board.jpg

Source: www.wevolver.com

https://www.unmannedsystemstechnology.com/wp-content/uploads/2025/02/accelerometers.jpg

Source: www.unmannedsystemstechnology.com

Thumbnail for easy-peasy.ai

https://media.easy-peasy.ai/8c791975-c28d-42fc-b4c7-5c37a2ed4357/1568d622-e862-4495-bbb0-732bc92a10fc_medium.webp

Source: easy-peasy.ai